Indian Institute of Technology, Kanpur

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Laboratory Report

RF Components: Characterization and Measurement (v2)

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Degree programme:	PhD
Submission date:	October 13, 2026

1 Objective

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or passband.

(b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

(c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

(d) To measure the power division and insertion loss of a power splitter/combiner.

(e) To measure amplifier gain, harmonics, the 1-dB compression point, and the third-order intercept point.

(f) To estimate the input reflection coefficient and return loss of the filter.

2 Theory

2.1 RF components in an RF receiver chain

A minimal RF receiver chain consists of several key components. A filter selects or rejects specified frequency ranges. A mixer translates a signal from one frequency to another. A low-noise amplifier (LNA) amplifies weak received signals while adding minimal noise. A power splitter/combiner distributes a signal to multiple paths or combines signals from different paths.

2.2 Filter transmission characteristics

A filter passes selected frequency components and attenuates others. The four common types are low-pass, high-pass, band-pass, and band-reject (notch) filters. For a transmission measurement, if P_{in} is the power measured without the filter and P_{out} is the power measured after inserting the filter, the insertion loss is defined as:

$$\text{IL} = P_{in} - P_{out} \text{ (dB)}$$

A smaller value of IL indicates better transmission through the filter. The 3 dB cutoff frequencies are measured relative to the minimum insertion loss in the passband.

2.3 Filter reflection and return loss

When the input impedance of the filter (Z_{in}) is not equal to the characteristic impedance of the RF system (Z_0), a part of the incident signal is reflected. The voltage reflection coefficient magnitude indicates the fraction of the incident voltage amplitude that is reflected:

$$|\Gamma| = \frac{V_{ref}}{V_{inc}} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}$$

The return loss is then given by:

$$\text{RL} = -20 \log_{10} |\Gamma| \text{ (dB)}$$

2.4 Quarter-wave open-circuit stub filter

An open-ended coaxial stub connected in shunt through a Tee connector behaves as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end is transformed approximately into a low impedance at the Tee junction, producing a notch. The physical length of the stub is approximately:

$$L = \frac{\lambda}{4} = \frac{VF \cdot c}{4 f_0}$$

where L is the stub length, VF is the velocity factor, c is the speed of light, and f_0 is the expected notch frequency.

2.5 Mixer

An RF mixer translates a signal from one frequency to another. When signals at f_{RF} and f_{LO} are applied, the mixer produces sum and difference components:

$$f_{IF} = f_{RF} + f_{LO} \quad \text{and} \quad f_{IF} = f_{RF} - f_{LO}$$

The conversion loss is calculated as the difference between the RF input power and the selected IF output power:

$$\text{Conversion loss} = P_{RF} - P_{IF} \text{ (dB)}$$

2.6 Power splitter/combiner

A power splitter divides the input power between two or more output ports. For an ideal two-way equal power splitter, each output is 3 dB below the input. The losses to the two output ports are calculated as:

$$Loss_1 = P_{in} - P_{out1} \quad \text{and} \quad Loss_2 = P_{in} - P_{out2}$$

2.7 RF amplifier gain, harmonics, and TOI

An RF amplifier increases the power of an input signal, with small-signal gain given by $Gain = P_{out} - P_{in}$. At sufficiently high input power, the gain decreases; the 1-dB compression point is the value of P_{in} where gain falls by 1 dB.

When two input tones at f_1 and f_2 are applied to a nonlinear amplifier, the output contains fundamentals, harmonics, and third-order intermodulation products ($2f_1 - f_2$ and $2f_2 - f_1$). The fundamental output increases with a 1:1 slope, while the third-order products increase with a 3:1 slope. Their extrapolated intersection is the third-order intercept (TOI).

3 Components Required

RF signal generator(s), spectrum analyser, digital storage oscilloscope (DSO), filter, mixer, power splitter/combiner, RF amplifier and required DC power supply, RF Tee connector, open-ended coaxial stub, short reference coaxial cable, 50 Ω terminations, and suitable RF cables, adapters, and attenuators.

4 Experimental Setup and Procedure

- **Filter characterization:** The RF signal generator was connected to the spectrum analyzer to record P_{in}, then the filter was inserted to record P_{out} across a frequency range to determine insertion loss and filter type. To measure return loss, a Tee connector and oscilloscope were used to record V_{max} and V_{min} over a small frequency range in the passband.
- **Stub filter:** An open-ended coaxial stub was connected to a Tee connector. P_{in} and P_{out} were measured to calculate attenuation and identify the notch frequency.
- **Mixer characterization:** RF and LO signals were applied to the mixer, and the spectrum analyzer was used to measure the IF output frequency and power to calculate conversion loss.
- **Power splitter/combiner:** Input power and output powers at the splitter ports were measured to calculate power division and insertion loss.
- **Amplifier characterization:** Small-signal gain was measured across the operating band. Input power was incrementally increased to plot the linearity curve and identify the 1-dB compression point. Finally, two closely spaced tones were applied using a combiner to measure fundamental and third-order products for estimating the Third-Order Intercept (TOI) point.

5 Observations and Results

5.1 Filter Transmission Response and Insertion Loss

Input Frequency (MHz)	P_{in} without filter (dBm)	P_{out} with filter (dBm)	IL (dB)
50	-0.91	-1.18	0.27
100	-0.80	-1.23	0.43
150	-0.92	-1.31	0.39
200	-1.09	-1.42	0.33
250	-0.95	-1.84	0.89
300	-1.17	-2.06	0.89
350	-1.37	-1.78	0.41
400	-1.12	-2.42	1.30
450	-1.22	-2.69	1.47
500	-1.53	-3.17	1.64
550	-1.43	-4.09	2.66
600	-1.42	-8.13	6.71
650	-1.48	-20.21	18.73
700	-1.24	-30.95	29.71

Table 1: *Filter transmission and insertion loss measurements*

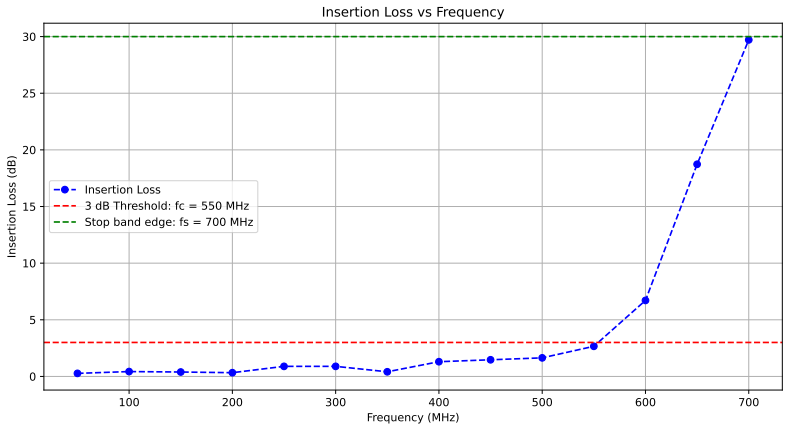


Figure 1: *Insertion loss versus frequency graph of the given filter.*

5.2 Filter Input Reflection Coefficient and Return Loss

Frequency (MHz)	V_{max} **(mV)**	V_{min} **(mV)**	V_{inc} **(mV)**	V_{ref} **(mV)**	**RL (dB)**
139 – 261	164	136	150	14	-20.63

Table 2: *Filter reflection and return loss measurements*

5.3 Quarter-Wave Open-Circuit Stub Notch Filter

For an open-ended coaxial stub made of standard polyethylene dielectric cable (velocity factor $VF = 0.66$) with a measured physical length of $L = 61.5$ cm, the theoretical expected notch frequency was calculated as:

$$f_0 = \frac{VF \cdot c}{4L} = \frac{0.66 \times 3 \times 10^8 \text{ m/s}}{4 \times 0.615 \text{ m}} \approx 80.48 \text{ MHz}$$

Frequency (MHz)	P_{in} **without stub (dBm)**	P_{out} **with stub (dBm)**	**Attenuation (dB)**
50	-0.71	-2.30	1.59
60	-0.72	-3.64	2.92
70	-0.50	-8.99	8.49
75	-0.48	-15.25	14.77
80	-0.50	-28.50	28.00
85	-0.56	-17.30	16.74
90	-0.51	-11.40	10.89
95	-0.53	-8.26	7.73
100	-0.60	-5.70	5.10
105	-0.70	-4.07	3.37
110	-0.83	-2.48	1.65

Table 3: *Stub notch filter attenuation measurements*

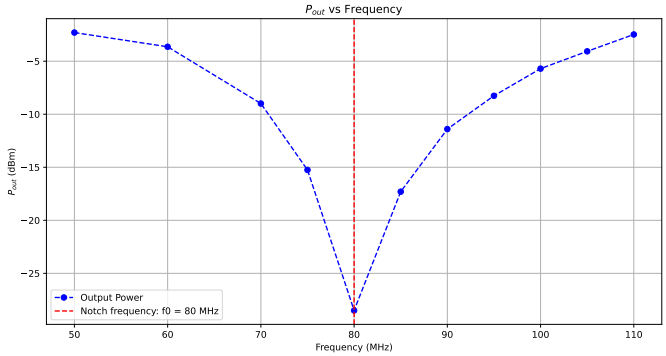


Figure 2: *Output characteristics of a notch filter*

5.4 Mixer Characterization

f_{RF} (MHz)	f_{LO} (MHz)	Expected f_{IF} (MHz)	Measured f_{IF} (MHz)	P_{IF} (dBm)	Conversion Loss (dB)
100	10	110 (Sum)	110	-9.62	9.62
		90 (Diff)	90	-9.56	9.56
110	10	120 (Sum)	120	-9.56	9.56
		100 (Diff)	100	-9.56	9.56
120	10	130 (Sum)	130	-9.48	9.48
		110 (Diff)	110	-9.56	9.56
130	10	140 (Sum)	140	-9.46	9.46
		120 (Diff)	120	-9.28	9.28

Table 4: *Mixer frequency conversion and loss measurements (P_{RF} = 0 dBm)*

5.5 Power Splitter/Combiner

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	$Loss_1$ (dB)	$Loss_2$ (dB)
50	-1.07	-4.42	-4.42	3.35	3.35

Table 5: *Power splitter insertion loss and power division*

Condition	f_1 (MHz)	P_1 (dBm)	f_2 (MHz)	P_2 (dBm)	Measured Output
Equal Frequencies	50	0	50	0	50 MHz @ -1.51 dBm
Different Frequencies	50	-1.6	10	-1.6	50 MHz @ -2.87 dBm 10 MHz @ -1.64 dBm

Table 6: *Power combiner outputs for equal and different frequencies*

5.6 Amplifier Characterization

The RF amplifier was characterized for small-signal gain, linearity, and third-order intermodulation products. Table 7 and Figure 4 present the input-output power sweep. Note that the amplifier remained in its linear region, and the 1-dB compression point was not reached. Table 8 and Figure 5 detail the two-tone measurement used to extrapolate the Third-Order Intercept (TOI) point.

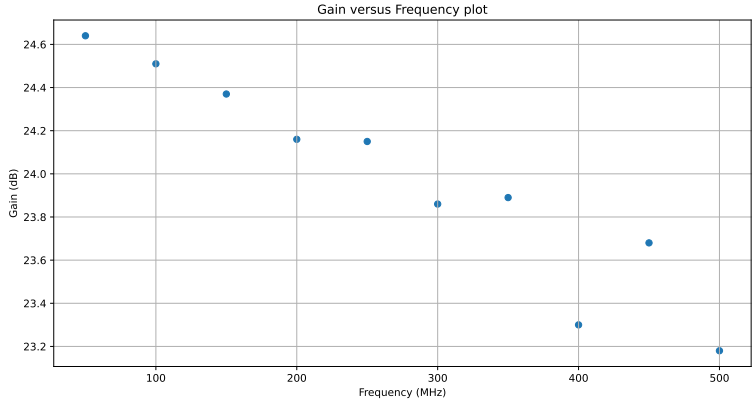


Figure 3: *Gain versus frequency response of the RF amplifier.*

P_{in} **(dBm)**	P_{out} **(dBm)**	**Gain (dB)**
-20.00	4.62	24.62
-19.90	4.70	24.60
-19.80	4.80	24.60
-19.70	4.90	24.60
-19.60	5.00	24.60
-19.50	5.10	24.60
-19.40	5.20	24.60
-19.30	5.30	24.60
-19.20	5.40	24.60
-19.10	5.49	24.59
-19.00	5.57	24.57
-18.95	5.62	24.57

Table 7: *Amplifier gain and input power sweep (linear region)*

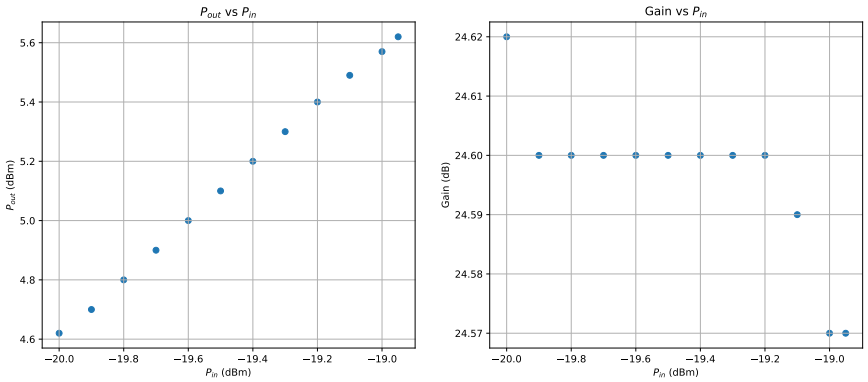


Figure 4: *Amplifier output power and gain versus input power.*

P_{in} **per tone (dBm)**	**Fundamental** P_{out} **(dBm)**	$P_{3rd-order}$ **(dBm)**
-33	-12.06	-73.74
-32	-11.18	-71.00
-31	-10.18	-68.50
-30	-9.20	-65.50
-29	-8.20	-62.50
-28	-7.17	-59.50

Table 8: *Amplifier intermodulation products for TOI estimation*

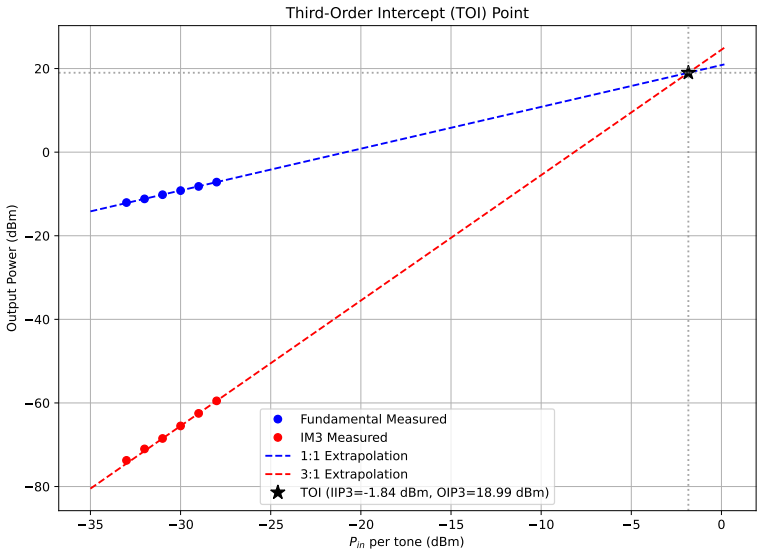


Figure 5: *Extrapolation of fundamental and IM3 products for TOI estimation (IIP3 = -1.84 dBm, OIP3 = 18.99 dBm).*

6 Results and Conclusion

Table 9 summarizes the principal measured and calculated values obtained across all characterized RF components, as required by the laboratory procedure.

Quantity	**Symbol**	**Measured/Calculated Value**
Low-pass/high-pass cutoff and stopband frequencies, if applicable	f_c, f_s	Low-pass: $f_c \approx 550$ MHz, $f_s = 700$ MHz
Band-pass/notch frequencies, if applicable	f_1, f_0, f_2	N/A (Tested filter was low-pass)
Filter insertion loss	IL	2.66 dB (at cutoff), < 1.64 dB in passband
Filter reflection coefficient	$\lvert\Gamma\rvert$	0.093
Filter return loss	RL	20.63 dB
Calculated and measured stub notch frequency	f_0	Calculated: 80.48 MHz Measured: 80.00 MHz
Mixer RF, LO, and IF frequencies	f_{RF}, f_{LO}, f_{IF}	f_{RF}: 100–130 MHz f_{LO}: 10 MHz f_{IF}: 90–140 MHz (Sum and Difference)
Power-splitter output powers	P_{out1}, P_{out2}	$P_{out1} = -4.42$ dBm $P_{out2} = -4.42$ dBm (for -1.07 dBm input)
Amplifier input power, output power, and gain	$P_{in}, P_{out}, Gain$	P_{in} range: -20.00 to -18.95 dBm P_{out} range: 4.62 to 5.62 dBm Small-signal Gain $\approx$ 24.6 dB
Third-order intercept point	TOI	IIP3 = -1.84 dBm OIP3 = 18.99 dBm

Table 9: *Summary of principal RF component measurements*

The RF components characterization experiment successfully evaluated the fundamental properties of a filter, mixer, power splitter, and open-circuit stub. The key findings are summarized below:

- **Filter Characteristics:** The component was identified as a low-pass filter. Based on the transmission response, the 3-dB cutoff frequency (f_c) was determined to be approximately 550

MHz (measuring 2.66 dB insertion loss). The stopband edge (f_s) was located at 700 MHz, providing 29.71 dB of signal rejection. Using the Tee-connector method within the passband, the input voltage reflection coefficient ($|\Gamma|$) was calculated to be 0.093. This equates to a return loss of 20.63 dB, indicating a strong impedance match with minimal signal reflection at the filter's input.

- **Quarter-Wave Stub:** The quarter-wave open-circuit stub successfully functioned as a notch filter. The measured notch frequency was clearly identified at 80 MHz, where the signal experienced a maximum attenuation of 28.00 dB. This closely matches the theoretical expected frequency of 80.48 MHz calculated from the cable's physical length and velocity factor.

- **Mixer Performance:** The RF mixer accurately produced the expected sum and difference intermediate frequencies (f_{IF}) for all tested inputs. The conversion loss remained highly consistent across the 100–130 MHz RF sweep, averaging approximately 9.5 dB. It should be noted that this conversion loss is slightly higher than ideal due to a power mismatch at the Local Oscillator (LO) port; the mixer requires a standard +7 dBm LO drive, which could not be perfectly matched during the experiment due to equipment limitations.

- **Power Splitter/Combiner:** The power splitter demonstrated balanced power division, measuring a 3.35 dB loss at both output ports for a 50 MHz input. This indicates a minimal excess insertion loss of only 0.35 dB beyond the ideal 3 dB theoretical split. The combiner also successfully multiplexed both identical and distinct frequency tones.

- **Amplifier Characterization:** Over the tested input power range of -20 dBm to -18.95 dBm, the amplifier operated strictly within its linear region, maintaining a relatively constant small-signal gain of approximately 24.6 dB. The 1-dB compression point was not reached, as the input power was not increased sufficiently to drive the amplifier into saturation (which would require observing the gain drop to roughly 23.6 dB). Additionally, a two-tone measurement successfully characterized the amplifier's non-linear distortion, yielding an extrapolated Input Third-Order Intercept (IIP3) of -1.84 dBm and an Output Third-Order Intercept (OIP3) of 18.99 dBm.

- **Measurement Uncertainty:** Minor deviations between theoretical expectations and experimental data are primarily attributed to uncalibrated coaxial cable and adapter insertion losses. Furthermore, visual estimation errors when reading voltage amplitudes directly from the oscilloscope, combined with the approximate nature of the uncalibrated Tee-junction setup, contributed to standard margins of error in the reflection coefficient calculations. Additionally, the unavailability of equipment to provide the exact +7 dBm input power required for the mixer's LO port resulted in sub-optimal diode switching, introducing a power mismatch that elevated the measured conversion loss.